



DIGITAL IMAGE FUNDAMENTALS

**Understanding Pixel Representation, Spatial Resolution,
Intensity Resolution, and Grayscale Images**

TECHNICAL PRESENTATION

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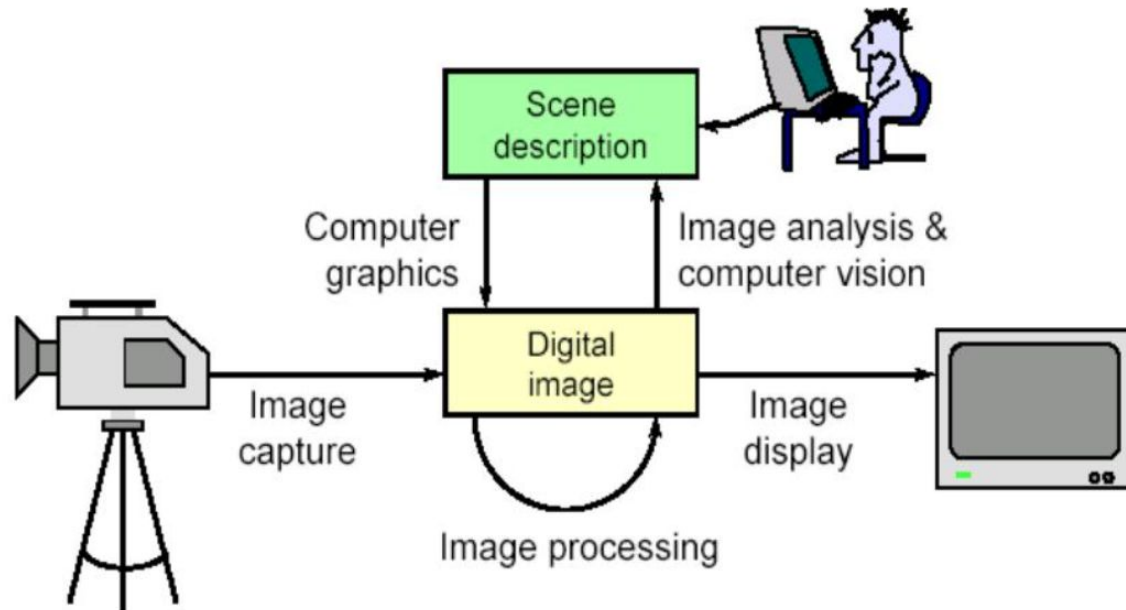
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INTRODUCTION



- - A digital image is a visual representation stored in digital form.
- - It consists of pixels arranged in rows and columns.
- - Used in photography, medical imaging, satellite imaging, and computer vision.
- - Image quality depends on resolution and intensity levels.





OBJECTIVES

Digital image = a multidimensional array of numbers (such as intensity image) or vectors (such as color image)

Each component in the image called pixel associates with the pixel value (a single number in the case of intensity images or a vector in the case of color images).

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the pixel value (a single number in
called pixel associates with
each component in the image

- - Understand digital image fundamentals.
- - Explain pixel representation.
- - Study spatial and intensity resolution.
- - Understand grayscale images.
- - Identify common image quality issues.

10	10	16	28
9	65	70	56
15	32	99	70
32	21	60	90
	85	85	43
	54	32	65

35	92	81	88
24	82	82	43
31	60	80	80
12	35	88	30
8	92	30	30
10	80	80	80



PROBLEM STATEMENT

- - Low spatial resolution causes blurred images.
- - Insufficient intensity resolution reduces detail.
- - Noise and distortion affect clarity.
- - Poor image quality impacts analysis and decision making.



Problem: How can image quality be represented and maintained effectively



DIGITAL IMAGE FUNDAMENTALS

- Pixel Representation: Smallest unit of an image.
- Spatial Resolution: Number of pixels in an image.
- Example: 1920×1080 .
- Intensity Resolution: Number of brightness levels.
- Example: 8-bit = 256 gray levels.
- Grayscale Image: Values range from 0 (black) to 255 (white).

- Picture elements (pixels)
 - Sampling, quantization
- Higher dimensional image -- voxels
- Bi-level images (black/0 or white/1)
- Grayscale images
 - 1 byte/pixel: 256 gray levels
- Color images
 - True color: RGB 24bits/pixel
- Image size, e.g. VGA 640×480
 - Grayscale image: 307,200 bytes
 - True color image: 921,600 bytes





CONCLUSION



- - Pixels form digital images.
- - Spatial resolution determines image detail.
- - Intensity resolution determines brightness accuracy.
- - Grayscale images are important in image processing.
- - These concepts are fundamental to computer vision.

*Thank
You!*